CYBORG SUPERMARKET ENTERPRISE DOCUMENTATION

Complete Project Handbook

Comprehensive Ecosystem Blueprint & Operational Guide

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Executive Publication Record

Organization:	Cyborg Omnichannel Engineering Organization
Ecosystem:	Windows POS • Android Mobile • Web Storefront • Admin Web • Backend
Primary Authority:	cyborg_system_architecture.pdf (16-Page Blueprint)
Source Dataset:	super_market(1).xlsx (1,000 Verified Egyptian Retail Records)
Security Level:	Enterprise Retail Distribution / Stakeholder Review
System Stack:	NestJS • TypeScript • PostgreSQL 16 • Redis 7 • SQLite • Flutter
Target Currency:	Egyptian Pounds (EGP) • Cairo Timezone (UTC+2 / UTC+3)

1. Executive Summary

Cyborg Supermarket represents an omnichannel retail ecosystem built specifically for the Egyptian supermarket industry. The platform bridges physical store branches and digital shopping channels through a unified product catalog, synchronized inventory balances, and centralized business logic.

> **[MISSION] Core Project Mission**
>
> Deliver an enterprise-grade retail platform connecting cashier terminals, mobile shoppers, onli

2. Business Problem & Market Opportunity

Traditional Egyptian grocery retail suffers from severe disconnected silos between POS cash registers and online e-commerce channels. Out-of-stock cancellations exceed 18% in legacy stores due to lagging stock updates. Cyborg solves this through real-time bidirectional synchronization, optimistic concurrency locking, and colloquial Egyptian Arabic voice shopping.

3. Ecosystem Component Inventory

Component	Platform / Tech	Primary Purpose	Network Mode
Windows POS Terminal	Flutter Desktop / Dart	High-speed cashier scanning &	Offline-First / Cloud Sync
Customer Mobile App	Flutter (Android / iOS)	Customer grocery shopping, fam	Online / Local Cache
Customer Storefront	Next.js / Responsive Web	Bilingual web shopping experie	Online REST API
Admin Dashboard	Next.js / Enterprise Web	Management overview, inventory	Online REST API
Backend API Gateway	NestJS / TypeScript	Clean Architecture modular mon	Server Runtime
Database & Cache	PostgreSQL 16 + Redis 7	Relational ACID persistence, p	High-Availability Cluster

4. User Roles & Stakeholder Access Matrix

Security enforces the principle of least privilege through hierarchical Role-Based Access Control (RBAC):

Role Code	Role Title	Scope of Authorization	Permitted Interfaces
SUPER_ADMIN	Super Administrator	Full system configuration, rol	Admin Web, CLI, API
STORE_MANAGER	Store Manager	Shift audits, refunds, price o	Admin Web, POS Supervisor
CASHIER	Checkout Cashier	Barcode scanning, cash tenderi	Windows POS Desktop
INVENTORY_OP	Inventory Operator	Stock receiving, damaged goods	Admin Web, Mobile Handheld
CUSTOMER	Shopper / Client	Product discovery, personal ca	Mobile App, Storefront
AUDITOR	Financial Auditor	Read-only access to sales, led	Admin Web (Read-Only)

5. Product Catalog & Verified Source Dataset

The initial product catalog is seeded from the verified real Egyptian supermarket dataset (super_market(1).xlsx), comprising exactly 1,000 active retail stock-keeping units (SKUs) across 5 major categories and 143 domestic brands.

- Currency Standard: All monetary amounts are denoted in Egyptian Pounds (EGP) with exact two-decimal precision.

- Bilingual Support: Every product maintains verified Arabic and English product names and categorization.

- Inventory Allocation: Initial inventory levels are allocated per category with automatic low-stock alerting.